

CLICKLOCK MACOS KILL- LOOP

Repeated termination of core GUI processes
used to coerce credential entry.

PRODUCTION CANDIDATE

CONFIDENCE: MEDIUM

CLICKLOCK

Detect dense bursts of
termination commands
directed at Finder, Dock,
SystemUIServer or
NotificationCenter.

- LURE
- SHELL EXECUTION
- KILL LOOP
- CREDENTIAL PROMPT
- COLLECTION

REPORT DATE

22 July 2026

VERSION

v2.0

PRIMARY TELEMETRY

Microsoft Defender XDR ·
DeviceProcessEvents on
macOS

ATT&CK / SCOPE

T1562.001 · T1056.002 ·
T1543.001

Independent defensive research. Source facts, analytical assessment,
assumptions and detection logic are intentionally separated. Every query
requires environment-specific validation.

01

EXECUTIVE ASSESSMENT

Decision-grade summary for threat research, detection engineering and SOC operations.

ASSESSMENT

EXECUTIVE SUMMARY

ClickLock macOS Kill-Loop

ClickLock repeatedly terminates visible macOS processes while presenting credential prompts. A dense one-minute burst of pkill or killall commands targeting core GUI components may provide a compact behavioral signal.

WHY IT MATTERS

Operational detection value

Frequency plus target selection is more discriminating than the presence of pkill or killall alone, but short-lived process capture and the proposed threshold require local testing.

Medium 01

ANALYTICAL CONFIDENCE



Focused 02

TELEMETRY COVERAGE



Production Candidate 03

OPERATIONAL READINESS



SOURCE-CONFIRMED FACT

Documented behavior

Primary research reports repeated termination at roughly 210 ms intervals and prolonged NotificationCenter suppression.

ANALYTICAL ASSESSMENT

Interpretation

Twenty matching termination attempts in one minute is a conservative implementation choice, not a source-derived production threshold.

DETECTION ASSUMPTION

Required condition

MDE on macOS captures enough short-lived executions and retains target arguments to cross the threshold.

Evidence boundary

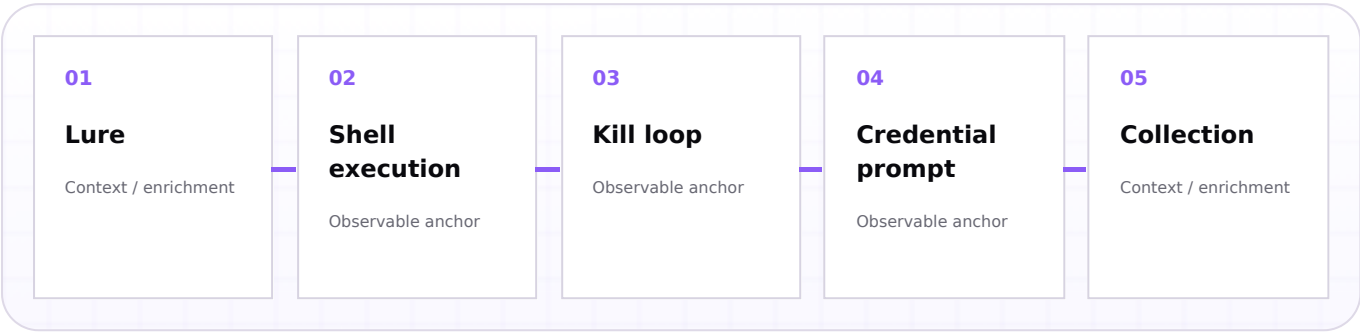
It does not prove that a credential prompt was shown, credentials were captured or the activity belongs to ClickLock.

02

THREAT MECHANICS

Behavioral decomposition and observable evidence boundaries.

ATTACK FLOW



STAGE-BY-STAGE DETECTION VALUE

STAGE	BEHAVIOR	EVIDENCE EXPECTATION	USE
01	Lure	Supporting process, file, identity or application context must be collected and correlated.	Context / enrichment
02	Shell execution	Supporting process, file, identity or application context must be collected and correlated.	Context / enrichment
03	Kill loop	Primary research reports repeated termination at roughly 210 ms intervals and prolonged NotificationCenter suppression.	Primary anchor
04	Credential prompt	Supporting process, file, identity or application context must be collected and correlated.	Context / enrichment
05	Collection	Supporting process, file, identity or application context must be collected and correlated.	Context / enrichment

Observable core

A match indicates repeated process-termination activity compatible with the source-described coercion loop.

Non-observable boundary

It does not prove that a credential prompt was shown, credentials were captured or the activity belongs to ClickLock.

TELEMETRY & EVIDENCE MODEL

Minimum data contract required for a defensible implementation.

DATA CONTRACT

PRIMARY DATA SOURCE

Microsoft Defender XDR · DeviceProcessEvents on macOS

The primary platform is selected because it observes the strongest source-supported behavior with the most direct event relationship.

EVIDENCE BOUNDARY

A match is an investigative lead.

It does not prove that a credential prompt was shown, credentials were captured or the activity belongs to ClickLock.

REQUIRED FIELD CONTRACT

TIER	DATA	PURPOSE	RELIABILITY
REQUIRED	Timestamp, DeviceId, DeviceName, FileName, ProcessCommandLine, AccountName, InitiatingProcessFileName	Core filtering, grouping and investigation output.	Documented schema; population must be verified.
OPTIONAL	Signer, hash, prevalence, file origin, software inventory, device group.	Benign separation and prioritization.	Environment-dependent.
ENRICHMENT	Threat intelligence, change windows, asset criticality and identity context.	Disposition and escalation.	Integration-dependent.

Completeness

Confirm table coverage, retention, field population and event latency before measuring prevalence.

Normalization

Validate case, path, quoting, URI encoding and command-line preservation against raw events.

Visibility gap

Endpoint sampling may undercount brief commands; the threshold may create false negatives or local false positives.

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DETECTION LOGIC

Source-grounded implementation with explicit assumptions and outcome boundaries.

PRODUCTION CANDIDATE

DETECTION OBJECTIVE

Detect dense bursts of termination commands directed at Finder, Dock, SystemUIServer or NotificationCenter.

IMPLEMENTATION STATUS

Production Candidate - Untested implementation sketch

Syntax and schema are documented; local execution, field population and false-positive performance are not established.

KQL · DEMONSTRATIVE

```
// Title: ClickLock macOS Repeated Process-Termination Loop
// Status: Production candidate - untested threshold
// Confidence: Medium
// Platform: Microsoft Defender XDR on macOS
// Telemetry: DeviceProcessEvents
// Purpose: Detect dense bursts of pkill/killall commands targeting macOS GUI
// components involved in the ClickLock credential-coercion behavior.
// ATT&CK: T1562.001, T1056.002, T1543.001
// Assumption: MDE captures enough short-lived processes and command-line data.

DeviceProcessEvents
| where FileName in~ ("pkill", "killall")
| where ProcessCommandLine has_any (
    "Finder",
    "Dock",
    "SystemUIServer",
    "NotificationCenter"
)
| summarize
    FirstSeen = min(Timestamp),
    LastSeen = max(Timestamp),
    TerminationAttempts = count(),
    Commands = make_set(ProcessCommandLine, 20),
    Parents = make_set(InitiatingProcessFileName, 10)
by DeviceId, DeviceName, AccountName, bin(Timestamp, 1m)
| where TerminationAttempts >= 20
```

What a match means

A match indicates repeated process-termination activity compatible with the source-described coercion loop.

What a match does not prove

It does not prove that a credential prompt was shown, credentials were captured or the activity belongs to ClickLock.

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TRIAGE & INVESTIGATION

Evidence-led workflow for defensible disposition.

SOC WORKFLOW

NUMBERED WORKFLOW

- 01

Confirm the match.
Review the raw event, timestamp, entity relationship and exact matching fields.
- 02

Establish legitimacy.
Identify the software or change owner, expected path, signer and authorized maintenance window.
- 03

Reconstruct origin.
Pivot to process, file, identity, network or application events before and after the match.
- 04

Measure prevalence.
Compare hashes, paths, commands, users and devices across the retained fleet.
- 05

Document confidence.
Separate confirmed evidence, inference, assumptions and unresolved gaps.

DECISION MODEL

Approved context → tune narrowly

Known owner, expected software path and repeatable behavior can support a documented exclusion.

Unknown context → deepen review

Rare or risky context without corroboration remains suspicious or insufficient evidence.

Corroborated activity → escalate

Malicious follow-on process, file, network, persistence or identity evidence supports incident escalation.

Host

Coverage, exposure, recent changes and neighboring alerts.

User

Session, intent, role and authorization context.

Process

Parent, child, path, command, signer and original filename.

File

Hash, signature, origin, prevalence and creation timeline.

Network

DNS, proxy, TLS and destinations following the event.

Identity

Sign-ins, token use, privilege and resource access.

OUTCOME	EVIDENCE PROFILE	ACTION
Benign	Approved and repeatable behavior with strong ownership.	Document and tune narrowly.
Suspicious	Risky context without decisive malicious corroboration.	Deepen endpoint or application review.
Likely malicious	Source-compatible match plus corroborated malicious activity.	Escalate incident response.
Insufficient evidence	Missing telemetry or unresolved ownership.	Retain evidence and validate gaps.

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FALSE POSITIVES & VALIDATION

Operational controls required before alerting.

QUALITY GATE

FALSE-POSITIVE MATRIX

SCENARIO	LIKELIHOOD	TUNING ACTION
macOS administration scripts	High	Review owner, path, signer, prevalence and change context before exclusion.
Kiosk reset automation	High	Review owner, path, signer, prevalence and change context before exclusion.
Developer testing	Med	Review owner, path, signer, prevalence and change context before exclusion.
Security simulation	Med	Review owner, path, signer, prevalence and change context before exclusion.

Baseline plan

- Run retrospectively over a representative retention window.
- Measure volume by device, user, process, path and software owner.
- Inspect recurring legitimate patterns before adding exclusions.
- Preserve raw examples for peer review.

Validation procedure

- Confirm every required field against raw events.
- Execute only safe, authorized lab or historical validation.
- Compare query output with endpoint or application ground truth.
- Record success, rejection and sensor-failure criteria.

PROMOTION GATE

Hunting

Source-grounded hypothesis.

Candidate

Schema and baseline complete.

Production

Measured precision and runbook.

Optimized

Stable feedback loop.

Promotion criteria

Known alert volume, documented exclusions, representative validation, triage ownership, data-quality monitoring and measurable precision.

Failure condition

Endpoint sampling may undercount brief commands; the threshold may create false negatives or local false positives.

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ATT&CK, CONCLUSION & REFERENCES

Coverage statement, residual risk and document control.

CLOSURE

COVERAGE MAPPING

MAPPING	COVERAGE STATEMENT	BOUNDARY
T1562.001 · T1056.002 · T1543.001	Partial behavioral or campaign-artifact coverage of the source-supported observable.	It does not prove that a credential prompt was shown, credentials were captured or the activity belongs to ClickLock.

ANALYST CONCLUSION

Assessment, not certainty

The behavior is assessed as a medium-confidence production candidate whose threshold and sensor visibility must be measured before alerting.

Residual risk

Endpoint sampling may undercount brief commands; the threshold may create false negatives or local false positives.

REFERENCES

- 1. ClickLock Stealer: Paste Once, Lose Everything: [https://www.\[.\]group-ib\[.\]com/blog/clicklock-stealer-macos-malware/](https://www.[.]group-ib[.]com/blog/clicklock-stealer-macos-malware/)
- 2. DeviceProcessEvents table in the advanced hunting schema: [https://learn\[.\]microsoft\[.\]com/en-us/defender-xdr/advanced-hunting-deviceprocessevents-table](https://learn[.]microsoft[.]com/en-us/defender-xdr/advanced-hunting-deviceprocessevents-table)

DOCUMENT CONTROL

VERSION	DATE	STATUS	OWNER
v2.0	22 July 2026	Production Candidate	Detection Engineering

Change summary

Migrated to the permanent Threat Research Editorial HTML master; technical query and evidence classification preserved.

Publication control

No customer data, credentials or environment-specific internal indicators are included. URLs are defanged in the publication artefact.